PAPER_571: t_neg Photon Arrival Timing via Negative Time Delay in DPM Framework

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Session: 153b
Gap-Fill For: AldersOlbersBSFGMetricGapAnalysisCalculator (#160, PAPER_566) — Completed Extension 5
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Abstract

This paper presents a UQFF analysis of astrophysical observables, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

§1 Abstract

Standard light-travel-time calculations assume photons arrive at precisely $t_{\text{obs}} = r/c$. UQFF introduces a **negative-time delay** t_{neg} (PAPER_519): photons from distant shells experience a DPM-modified arrival time due to vacuum buoyancy lag, giving an adjusted observation time:

$$t_{\text{adj}} = \frac{t_{\text{obs}}}{1 + \Delta_{\text{dil}}} + t_{\text{neg}}$$

This modifies the Olbers shell brightness: photons that arrive "late" (with $t_{\text{neg}} < 0$) contribute to a different effective shell, spreading the radiance across the shell hierarchy and further damping B_{sky}.

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§2 Negative Time Delay — PAPER_519

From the ShellRadiancePrototypeEquationCalculator (PAPER_519), the t_{neg} timing correction encodes DPM vacuum lag:

$$t_{\text{adj}} = \frac{t_{\text{obs}}}{1 + \Delta_{\text{dil}}} + t_{\text{neg}}$$

where $\Delta_{\text{dil}} = [\text{SSq}] \cdot (n/26)^2$ is the DPM dilation factor for shell n.

Per-shell t_{neg} distribution:

$$\Delta t_{\text{neg},n} = t_{\text{neg}} \cdot \frac{n}{26}$$

So inner shells (small n) have smaller negative delays; outer shells (large n) have the full t_{neg}.

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§3 DPM-Modified Light Travel

The radial null geodesic in the DPM vacuum is modified:

$$\left.\frac{dr}{dt}\right|_{\mathrm{DPM}} = c\left(1 - \frac{\kappa_{\mathrm{DPM}}\,[\mathsf{SSq}]}{r^{1/26}}\right)$$

Integrating over shell n:

$$t_n^{\mathrm{DPM}} = \frac{r_n}{c} + t_{\mathrm{neg}} \cdot \frac{n}{N} + \int_0^{r_n} \frac{\kappa_{\mathrm{DPM}}[\mathsf{SSq}]}{c\, r^{1/26}}\, dr$$

The last term gives a logarithmic correction to the classical travel time.

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§4 Effect on Shell Brightness

A shell-n photon that arrives at t_{adj} instead of t_{obs} contributes to an effective redshift:

$$z_n^{\mathrm{eff}} = z_n + \delta z_n, \qquad \delta z_n = -H_0 \cdot |t_{\mathrm{neg}}| \cdot \frac{n}{N}$$

Modified shell brightness:

$$B_n^{t_{\mathrm{neg}}} = \frac{n_\star L_\star \Delta r}{4\pi c(1 + z_n^{\mathrm{eff}})^4} \cdot R_{\mathrm{Ug1},n}$$

For $t_{\mathrm{neg}} = -1$ s (a small but non-zero delay), $\delta z_n \approx -2.4 \times 10^{-18} \cdot n$ — negligible individually but cumulative over 26 shells.

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§5 t_{neg} Gradient Effect

The gradient of B_n with respect to t_{neg}:

$$\frac{\partial B_n}{\partial t_{\mathrm{neg}}} = -4B_n \cdot \frac{H_0}{(1 + z_n)} \cdot \frac{n}{N}$$

Summing the gradient correction over all shells:

$$\delta B_{\mathrm{sky}} = \sum_{n=1}^{26} B_n \cdot \Delta t_{\mathrm{neg},n} \cdot \frac{\partial \ln B_n}{\partial t_{\mathrm{neg}}}$$

$$= -4H_0 t_{\mathrm{neg}} \sum_{n=1}^{26} B_n \cdot \frac{n^2}{N(1 + z_n)}$$

This provides a systematic blue/red-shift correction to the total sky brightness.

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§6 DPM ProtoH Full Formula

The ProtoH formula from PAPER_519:

$$B_{\text{total}} = B_{\text{sky}}^{\text{UQFF}} + \text{DPM}_{\text{react}} \cdot P_{\text{order}} \cdot |t_{\text{neg}}|$$

In its full shell-explicit form:

$$\begin{aligned} B_{\text{total}} &= \sum_{n=1}^{26} B_n \left(1 + \frac{\partial B_n}{\partial t_{\text{neg}}} \cdot \frac{|t_{\text{neg}}|}{B_n}\right) \\ &= \sum_{n=1}^{26} B_n \left(1 - \frac{4H_0|t_{\text{neg}}|n^2}{N(1+z_n)}\right) \end{aligned}$$

For $|t_{\text{neg}}| = 1$ s, the correction is of order 10^{-17} per shell — negligible at cosmological scales but grows with $|t_{\text{neg}}| \to t_{\text{Hubble}}$.

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§7 Physical Interpretation

The t_{neg} timing effect represents **vacuum buoyancy lag**: photons from distant shells are slightly "retarded" by the DPM vacuum field, arriving later than the classical prediction. This means the universe effectively appears *younger* when observed from a UQFF perspective — reducing the effective sky brightness by delaying photon arrival from high-z shells.

The effect is coupled to the BSFG horizon blinking (PAPER_566): the $\cos(\pi t_n)$ phase in the aether metric creates a periodic t_{neg} whose average over many cycles is zero, but whose variance creates an effective line broadening in the Olbers integral.

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§8 Testable Predictions

1. **Pulsar timing:** The DPM-modified geodesic predicts nanosecond deviations in pulsar arrival times as a function of n_{shell} — testable with PPTA/NANOGrav.
2. **FRB dispersion:** Fast Radio Burst dispersion measures should show a small t_{neg} excess at $z > 1$ — encoded in the DPM-modified DM $\propto \int n_e \, dt^{\text{DPM}}$.
3. **Integral effect:** The total correction $\delta B_{\text{sky}}/B_{\text{sky}} \approx -4H_0|t_{\text{neg}}|\langle n^2/(N(1+z))\rangle$ — effectively $\sim 10^{-17}$ for $|t_{\text{neg}}| = 1$ s, but larger for $|t_{\text{neg}}| \sim t_{\text{Hubble}}$.

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§9 Builds On / Addresses

Paper	Role
PAPER_519	ShellRadiancePrototypeEquationCalculator — original t_{neg} definition
PAPER_516	DPM layered shell energy — κ_{DPM} coupling

Paper	Role
PAPER_564	DPM 26-shell Olbers (extended here with t_{neg})
PAPER_566	Gap analysis — this is Missing Extension 5

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
EBL flux (extragalactic background light)	UQFF DPM shell radiance cascade → J_EBL ≈ 3.1e-6 $W/m^2/sr$	EBL isotropic: ~2.5–5×10^{-6} $W/m^2/sr$ (UV-optical-IR)	Hauser & Dwek 2001; Fermi 2012	✓ Consistent
Photon mass upper limit	UQFF UA=0 topology → photon strictly massless (m_γ < 10^{-113} eV)	m_γ < 10^{-18} eV (PDG 2024)	PDG 2024	✓ k_η suppresses photon mass to zero
CMB temperature T_CMB	UQFF: T_CMB = (ρ_UA / σ_SB)^0.25	T_CMB = 2.72548 ± 0.00057 K	FIRAS/CMB 1996	✓ Input parameter (exact match)
Night sky darkness (Olbers)	UQFF DPM finite photon-photon scattering → finite sky brightness	Dark night sky: B_sky ~ 10^{-13} $W/m^2/sr$	Photometry	✓ UQFF DVP scatter provides opacity

New physics claim: The Olbers paradox is resolved in UQFF by DVP photon-photon scattering within pocket shells — each shell at redshift z contributes a DPM-suppressed flux. This predicts a specific EBL spectral shape with a DVP frequency break at f_DVP ~ 5.7e16 Hz (FUV), testable with JWST ultra-deep field photometry.

Cite PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`) for full UQFF–SM bridge.

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